Differentiation — Chain Rule Skill Ladder

ANSWER KEY

Chunk 0 — Special angle values

1. **0**

2. **√2/2**

3. **√3**

4. **1**

5. **1**

cos0=1, sec0=1/1=1, $\sec^2 0 = 1^2 = 1$

6. **√3/2**

7. **1**

8. **1**

9. **2**

cos(π/4)=√2/2, sec=2/√2=√2, $\sec^2 = 2$

10. **4**

cos(π/3)=1/2, sec=2, $\sec^2 = 4$

Chunk 1 — Basic derivative facts

1. $5x^4$
2. e^x
3. $1/x$
4. $\cos x$
5. $-\sin x$
6. $\sec^2 x$
7. 0
8. 1
9. $1/(2\sqrt{x})$
   $(=\frac{1}{2}x^{-1/2})$
10. $-1/x^2$
    $(=-x^{-2})$

Chunk 2 — Derivative of a straight line

1. 3
2. 7
3. −4
4. 1/2
5. −2
6. 5
7. 9
8. −1
9. 2
10. 10

Chunk 3 — Spot outer/inner (easy case)

1. **Outer: e^(box). Inner: 3x+1**
2. **Outer: sin(box). Inner: 2x−5**
3. **Outer: $(\text{box})^3$. Inner: 4x+7**
4. **Outer: ln(box). Inner: 5x+2**
5. **Outer: cos(box). Inner: 6x**
6. **Outer: $(\text{box})^5$. Inner: x+9**
7. **Outer: e^(box). Inner: −2x**
8. **Outer: tan(box). Inner: 3x+1**
9. **Outer: √(box) i.e. $(\text{box})^{1/2}$. Inner: 2x+7**
10. **Outer: $(\text{box})^4$. Inner: 10−x**

Chunk 4 — Chain rule, full computation (easy case)

1. **$dy/dx = 5e^{5x+2}$**

2. **$dy/dx = 4\cos(4x-1)$**

3. **$dy/dx = 12(3x+1)^3$**
   $4(3x+1)^3 \times 3$

4. **$dy/dx = 2/(2x+7)$**

5. **$dy/dx = -5\sin(5x)$**

6. **$dy/dx = 3(x-6)^2$**

7. **$dy/dx = -3e^{-3x}$**

8. **$dy/dx = 2\sec^2(2x+5)$**

9. **$dy/dx = 3/(2\sqrt{(3x+4)})$**
   $\frac{1}{2}(3x+4)^{-1/2} \times 3$

10. **$dy/dx = -10(8-2x)^4$**
    $5(8-2x)^4 \times (-2)$

Chunk 5 — Spot outer/inner (harder case)

1. **Outer: e^(box). Inner: x^2**
2. **Outer: sin(box). Inner: x^3**
3. **Outer: $(\text{box})^5$. Inner: x^2+1**
4. **Outer: ln(box). Inner: x^2+3**
5. **Outer: cos(box). Inner: $\sqrt{x}$**
6. **Outer: $(\text{box})^3$. Inner: $1/x + 1$**
7. **Outer: tan(box). Inner: x^2**
8. **Outer: e^(box). Inner: $4-x^2$**
9. **Outer: ln(box). Inner: $3x^3$**
10. **Outer: $\sqrt{}$(box). Inner: x^2+1**

Chunk 6 — Derivative of the inner piece alone

1. $2x$
2. $3x^2$
3. $2x$
4. $2x$
5. $1/(2\sqrt{x})$
6. $-1/x^2$
7. $-2x$
8. $9x^2$
9. $2x-5$
10. $-1/x^2$

Chunk 7 — Chain rule, full computation (harder case)

1. **$dy/dx = 2x \cdot e^{x^2}$**

2. **$dy/dx = 3x^2\cos(x^3)$**

3. **$dy/dx = 10x(x^2+1)^4$**
   $5(x^2+1)^4 \times 2x$

4. **$dy/dx = 2x/(x^2+3)$**

5. **$dy/dx = -\sin(\sqrt{x}) / (2\sqrt{x})$**
   $-\sin(\sqrt{x}) \times 1/(2\sqrt{x})$

6. **$dy/dx = 2x \cdot \sec^2(x^2)$**

7. **$dy/dx = -2x \cdot e^{4-x^2}$**

8. **$dy/dx = 3/x$**
   $(9x^2)/(3x^3)$ simplifies to $3/x$ — cross-check: $\ln(3x^3)=\ln 3+3\ln x$, derivative$=3/x$ ✓

9. **$dy/dx = x/\sqrt{(x^2+1)}$**
   $\frac{1}{2}(x^2+1)^{-1/2} \times 2x$

10. **$dy/dx = 4(2x-5)(x^2-5x)^3$**
    $4(x^2-5x)^3 \times (2x-5)$

Chunk 8 — Chain rule twice in a row

1. $\mathbf{dy/dx = 2\cos(2x)\cdot e^{\sin 2x}}$

2. $\mathbf{dy/dx = 3e^{3x}\cos(e^{3x})}$

3. $\mathbf{dy/dx = -5\tan(5x)}$

$(1/\cos 5x) \times (-5\sin 5x) = -5\sin 5x/\cos 5x = -5\tan 5x$

4. $\mathbf{dy/dx = 12\cos(3x)(\sin 3x)^3}$

$4(\sin 3x)^3 \times 3\cos 3x$

5. $\mathbf{dy/dx = -\sin(\ln 2x)/x}$

$-\sin(\ln 2x) \times$ (derivative of $\ln 2x = 1/x$)

6. $\mathbf{dy/dx = 2\cos(4x) / \sqrt{(\sin 4x)}}$

$\frac{1}{2}(\sin 4x)^{-1/2} \times 4\cos 4x$

7. $\mathbf{dy/dx = 6e^{2x}(e^{2x}+1)^2}$

$3(e^{2x}+1)^2 \times 2e^{2x}$

8. $\mathbf{dy/dx = 4\cot(4x)}$

$(1/\sin 4x) \times 4\cos 4x = 4\cos 4x/\sin 4x = 4\cot 4x$

9. $\mathbf{dy/dx = 4(2x+1)\cos((2x+1)^2)}$

$\cos((2x+1)^2) \times 2(2x+1)\times 2$

10. $\mathbf{dy/dx = -3\sin(3x)\cdot e^{\cos 3x}}$